Sharp six-color bound for 3-dynamic coloring of claw-free cubic graphs

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Allowed standard prerequisites (editorial).

- Dynamic colorings, claw-free cubic decompositions, incidence coloring and circulant graph squares. Native proper r-dynamic coloring and graph-square conventions, complete original reduction and coloring words, author-used external incidence choosability theorem/citation and all original TikZ figures retained.. All source-local proof support is present, including nonadjacent small-diamond appendix cases; no author claim is independently certified or reconstructed.

Abstract

An r-dynamic coloring of a graph G is a proper vertex coloring in which every vertex v sees at least $\min\{r, d(v)\}$ distinct colors in its neighborhood. The minimum number of colors in such a coloring is the r-dynamic chromatic number $\chi_r(G)$. We study dynamic colorings of regular graphs. A straighforward observation shows that $\chi_r(G) = \chi(G^2)$, for any r-regular graph G. We prove that $\chi_3(G) = \chi(G^2) \leq 6$ for every claw-free cubic graph G, and the bound is sharp. For Hamiltonian claw-free cubic graphs, we improve the bound to 5 apart from four explicit exceptions. We also determine exactly the r-dynamic chromatic number of the 4-regular circulant graph $C_p(1,3)$ for each $r \in \{2,3,4\}$. In particular, the case $r = 4$ determines $\chi(C_p^2(1,3))$.

1 Introduction

Throughout the paper, all graphs are finite, simple, and undirected, and we use standard terminology and notation [1, 6]. For a vertex v of a graph G, let $N_G(v)$ and $d_G(v)$ denote its neighborhood and degree, respectively. For a positive integer r, an *r-dynamic coloring* of a graph G is a proper vertex coloring c such that

$$|c(N_G(v))| \geq \min\{r, d_G(v)\}$$

for every $v \in V(G)$. The minimum number of colors in such a coloring is the *r-dynamic chromatic number* of G, denoted by $\chi_r(G)$. Dynamic coloring, corresponding to $r = 2$, was introduced by Montgomery [10]. Higher-order versions have subsequently been studied under the names conditional coloring, r-hued coloring, and r-dynamic coloring; see [3, 8].

Dynamic coloring provides a natural interpolation between ordinary coloring and coloring the square of a graph. Recall that G^2 has vertex set $V(G)$, with two distinct vertices adjacent whenever their distance in G is at most two. For every graph G,

$$\chi(G) = \chi_1(G) \leq \chi_2(G) \leq \cdots \leq \chi_{\Delta(G)}(G) = \chi(G^2). \tag{1}$$

In particular, if G is d-regular, then

$$\chi_d(G) = \chi(G^2). \tag{2}$$

Thus, on regular graphs, the last member of the dynamic-coloring hierarchy is precisely the classical square-coloring problem.

Both dynamic coloring and coloring graph squares have received considerable attention for graphs of bounded degree. We write $\mathrm{ch}(G)$ for the list chromatic number of G. Cranston and Kim [4] proved that every connected subcubic graph other than the Petersen graph satisfies $\mathrm{ch}(G^2) \leq 8$, and that the bound is best possible. For planar cubic graphs, Thomassen [12] proved the sharp bound $\chi(G^2) \leq 7$. Square coloring has also been studied under structural restrictions. In particular, Cames van Batenburg and Kang [13] proved that every claw-free graph with clique number three satisfies $\chi(G^2) \leq 10$, and further bounds for squares of claw-free graphs were obtained in [5].

The regular setting is also natural from the dynamic-coloring perspective. Montgomery conjectured that every regular graph satisfies $\chi_2(G) \leq \chi(G)+2$. Bowler et al. [2] disproved this conjecture by showing that for every integer $k \geq 2$ there exists a regular graph G with $\chi(G) = k$ and $\chi_2(G) = 2k$. Thus, regularity alone does not force the dynamic chromatic number to remain close to the ordinary chromatic number, motivating the study of structured classes of regular graphs.

Our first main result concerns claw-free cubic graphs. For such graphs, the 3-dynamic coloring problem is exactly the square-coloring problem by (2). Li and Lai [9] proved that every claw-free graph satisfies $\chi_3(G) \leq \max\{\chi(G) + 3, 7\}$, which yields the upper bound 7 for claw-free cubic graphs. We improve this bound to 6, and the improvement is best possible.

- If G is a claw-free cubic graph, then $\chi(G^2) = \chi_3(G) \leq 6$. Moreover, the bound is sharp. (Theorem 5).

Indeed, the triangular prism $C_3 \square K_2$ has diameter two, and hence its square is K_6. The proof of Theorem 5 combines the structure of claw-free cubic graphs with incidence colorings of subcubic multigraphs. We then exploit the structure more strongly in the Hamiltonian case and reduce the bound by one, apart from four explicit graphs (to be defined later).

- If G is a Hamiltonian claw-free cubic graph then $\chi(G^2) = \chi_3(G) \leq 5$, except for four particular graphs. (Theorem 8).

For completeness, observe the case $r = 2$ is immediate for claw-free cubic graphs. The neighborhood of every vertex contains an edge, and hence every proper coloring is automatically 2-dynamic. Consequently, $\chi_2(G) = \chi(G)$.

Our second direction is to determine the complete dynamic-coloring hierarchy for a natural family of 4-regular graphs. For positive integers $p, n_1, \ldots, n_k$, the circulant graph $C_p(n_1, \ldots, n_k)$ has vertex set $\mathbb{Z}_p$, with i adjacent to $i \pm n_j$ for every j. We focus on $C_p(1, 3)$ and determine $\chi_r(C_p(1, 3))$ for every $r \in \{2, 3, 4\}$.

The following hold (Theorems 9, 10, and 11):

- If $p \geq 6$, then
$$\chi_2(C_p(1,3)) = \begin{cases} 3, & 9 \mid p, \\ 4, & \text{otherwise.} \end{cases}$$
- If $p \geq 12$, then
$$\chi_3(C_p(1,3)) = \begin{cases} 4, & 12 \mid p, \\ 5, & \text{otherwise.} \end{cases}$$
- If $p \geq 12$, then
$$\chi_4(C_p(1,3)) = \begin{cases} 5, & 5 \mid p, \\ 8, & p = 16, \\ 7, & p \in \{13, 14, 19, 21, 26, 31\}, \\ 6, & \text{otherwise.} \end{cases}$$

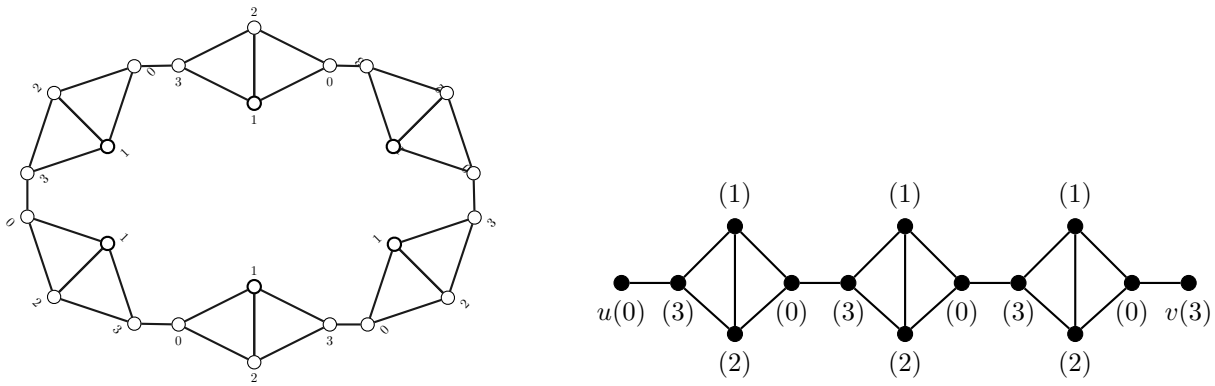
Since $C_p(1,3)$ is 4-regular for $p \geq 12$, the last part also determines the chromatic number of its square. In fact, $C_p^2(1,3) = C_p(1,2,3,4,6)$.

2 Claw-free cubic graphs

In this section we study 3-dynamic colorings of claw-free cubic graphs. A *diamond* is an induced subgraph isomorphic to $K_4 - e$. We say that two diamonds are *adjacent* if there exists an edge joining a degree-2 vertex of one diamond to a degree-2 vertex of the other. Contracting each diamond to a single vertex and preserving these adjacencies yields an auxiliary graph; if this graph is a path, we call the corresponding subgraph a *string of diamonds*, and if it is a cycle, we call it a *ring of diamonds*.

We have the next observation, whose proof is direct by inspection (Figure 1a).

Observation 1. *If G is a ring of diamonds then $\chi_3(G) \leq 5$.*



(a) (b)

Figure 1: A ring of diamonds (a) and a string of diamonds (b). In both cases, the assigned colors apear in parenthesis.

Proposition 2. *Let F be a graph with $\Delta(F) = 3$.*

1. *If G is obtained from F by replacing an edge uv with a string of diamonds, then $\chi_3^d(G) \leq \chi_3^d(F)$.*
2. *Let u and v be adjacent vertices of degree two in F having a common neighbor w. If G is obtained from F by adding a nonempty string of diamonds joining u and v, then $\chi_3^d(G) \leq \max\{\chi_3^d(F), 6\}$.*

Proof. Let ϕ be a 3-dynamic coloring of F.

For the first statement, since $\Delta(F) = 3$, at least four colors are available. Assume $\phi(u) = 0$ and $\phi(v) = 3$. For each diamond in the string replacing uv, color its two degree-3 vertices with 1 and 2, and its two degree-2 vertices with 3 and 0, oriented so that the vertex closer to u receives 3 and the vertex closer to v receives 0. (Figure 1b). Then every vertex of the string sees the required number of distinct colors. Moreover, at u the deleted neighbor v of color 3 is replaced by a neighbor of color 3, and similarly at v the deleted neighbor u of color 0 is replaced by a neighbor of color 0. Thus ϕ extends to a 3-dynamic coloring of G using no new colors, and $\chi_3^d(G) \leq \chi_3^d(F)$.

For the second statement, since u, v, w induce a triangle, we may assume $\phi(u) = 0$, $\phi(v) = 1$, $\phi(w) = 2$. Orient the string from u to v. In every diamond, color its two degree-3 vertices with 2 and 5, and its two degree-2 vertices with 3 and 4, with the vertex closer to u receiving 3 and the vertex closer to v receiving 4. Thus consecutive diamonds are joined by an edge whose endpoints have colors 4 and 3.

The three neighbors of u receive colors $1, 2, 3$, and the three neighbors of v receive colors $0, 2, 4$. Every internal vertex of the string also has three neighbors of pairwise distinct colors. All other neighborhoods are unchanged. Hence ϕ extends to a 3-dynamic coloring of G with at most $\max\{\chi_3^d(F), 6\}$ colors. □

Oum characterized bridgeless claw-free cubic graphs in terms of a reduction to cubic multigraphs. To handle graphs with bridges, we extend Proposition 1 of Oum [11] as follows.

Proposition 3. *Let G be a connected claw-free cubic graph. Then one of the following holds:*

1. *$G \cong K_4$;*
2. *G is a ring of diamonds;*
3. *G is obtained from a connected cubic multigraph H, possibly containing cutedges and loops, by replacing every vertex of H with a triangle and every edge of H with a possibly empty string of diamonds. Moreover, every loop of H is replaced with a nonempty string of diamonds.*

Proof. We use the elementary observations appearing in the proof of Proposition 1 of Oum. Every vertex of G belongs to a triangle. Moreover, if a vertex belongs to 3 triangles, then $G \cong K_4$. Thus, we may assume that every vertex belongs to at most 2 triangles. A vertex belonging to two triangles lies in an induced diamond, and distinct diamonds are vertex-disjoint. Every vertex not contained in a diamond consequently belongs to exactly one triangle. Hence, the vertex set of G has a partition $\mathcal{P}$ whose members induce triangles or diamonds.

Let B be the multigraph obtained by contracting every member of $\mathcal{P}$ to a single vertex. The edges of B correspond to edges of G joining distinct members of $\mathcal{P}$. Note that B is loopless but can have parallel edges. If $P \in \mathcal{P}$ is a triangle, then the corresponding vertex of B has degree three. If P is a diamond, then the corresponding vertex of B has degree two. Therefore, $d_B(v) \in \{2, 3\}$ for every $v \in V(B)$. If every vertex of B has degree two, then B is a cycle, since B is connected. In this case, G is a ring of diamonds. Otherwise, suppress all degree-two vertices of B. That is, every maximal path whose internal vertices have degree two is replaced with one edge. The two endvertices of such a path are allowed to coincide, in which case the resulting edge is a loop.

Let H be the resulting multigraph. Every vertex of H has degree three, and hence H is a connected cubic multigraph. (It may contain parallel edges, bridges and loops). Every edge e of H corresponds to a maximal path of B whose internal vertices represent diamonds. If this path has k internal vertices, then e corresponds to a string of k diamonds. Reversing the contractions shows that G is obtained from H by replacing its vertices with triangles and its edges with strings of diamonds. Moreover, if e is a loop of H, then its corresponding path in B begins and ends at the same triangle block. As B is loopless, the path contains at least one diamond block, and the loop is replaced with a nonempty string of diamonds. □

Finally, we need the following auxiliary result. An *incidence* of a loopless multigraph H is a pair (v, e), where $v \in V(H)$ and $e \in E(H)$ is incident with v. Two incidences (u, e) and (v, f) are said to be *adjacent* if at least one of the following conditions holds: $u = v$, $e = f$, or one of e and f is an edge joining u and v. An *incidence list assignment* of H is a function L that assigns to every incidence (v, e) a nonempty set $L(v, e)$ of colors. An *incidence L-coloring* is a function φ such that $\varphi(v, e) \in L(v, e)$ for every incidence (v, e), and adjacent incidences receive distinct colors.

The multigraph H is *incidence k-choosable* if it admits an incidence L-coloring for every incidence list assignment L satisfying $|L(v,e)| \geq k$ for every incidence (v,e). The *incidence choice number* of H, denoted by $\mathrm{ch}_i(H)$, is the least integer k for which H is incidence k-choosable.

We shall use the following theorem of Kang and Park.

Theorem 4 (Kang and Park [7])**.** *Every loopless subcubic multigraph H is incidence 6-choosable. Equivalently,* $\mathrm{ch}_i(H) \leq 6$.

We are ready to show our first theorem.

Theorem 5. *Let G be a claw-free cubic graph. Then $\chi_3(G) \leq 6$. Moreover, the bound is tight.*

Proof. It is enough to consider G connected. If $G \cong K_4$, the result is immediate. If G is a ring of diamonds, the result follows from Observation 1. Hence, by Proposition 3, we may assume that G is obtained from a connected cubic multigraph H by replacing its vertices with triangles and its edges with possibly empty strings of diamonds.

Remove all loops of H, and let $\widehat{H}$ be the resulting loopless subcubic multigraph. By Theorem 4, $\widehat{H}$ has an incidence coloring ϕ using at most six colors. Let $\widehat{G}$ be obtained from $\widehat{H}$ by replacing every vertex x with a triangle T_x. If $u \in T_x$ corresponds to an edge $e = xy \in E(\widehat{H})$, set

$$\widehat{\phi}(u) = \phi(x,e).$$

If $d_{\widehat{H}}(x) = 3$, then the three colors seen by any vertex of T_x correspond to three pairwise adjacent incidences of $\widehat{H}$, and hence are distinct. If $d_{\widehat{H}}(x) = 1$, then T_x contains two adjacent vertices of degree two. Let w be the third vertex of T_x and let z be its neighbor outside T_x. Since $\widehat{\phi}(w) \neq \widehat{\phi}(z)$, we can assign to the two degree-two vertices two distinct colors, both different from $\widehat{\phi}(w)$ and $\widehat{\phi}(z)$. Thus $\widehat{\phi}$ extends to a 3-dynamic coloring of $\widehat{G}$, and therefore $\chi_3(\widehat{G}) \leq 6$. Finally, G is obtained from $\widehat{G}$ by repeatedly applying the two operations of Proposition 2: non-loop edges are replaced by strings of diamonds, whereas each deleted loop is recovered by adding a string of diamonds between the two corresponding degree-2 vertices of the same triangle. Hence

$$\chi_3^d(G) \leq \chi_3^d(\widehat{G}) \leq 6.$$

The bound is tight. Indeed, the triangular prism $C_3 \square K_2$ is cubic and claw-free and has diameter two. In any 3-dynamic coloring, two adjacent vertices receive distinct colors, and two vertices with a common neighbor also receive distinct colors. Hence all six vertices of $C_3 \square K_2$ must receive distinct colors, and therefore $\chi_3(C_3 \square K_2) = 6$. □

We now proceed with the Hamiltonian case. By Proposition 3, a loop or a bridge of H would give rise to a bridge in G. Hence we have the next corollary.

Corollary 6. *Let G be a connected bridgeless claw-free cubic graph which is neither K_4 nor a ring of diamonds. Then G is obtained from a loopless bridgeless cubic multigraph H by replacing every vertex of H with a triangle and every edge of H with a possibly empty string of diamonds.*

Let X_{12} be the graph obtained from the cubic multigraph consisting of a 4-cycle with two opposite edges doubled by replacing each vertex with a triangle. Let X_{16}^1 and X_{16}^2 be the graphs obtained from X_{12} by inserting one diamond into, respectively, an edge belonging to a parallel pair and a simple edge of the underlying cubic multigraph. We need the next technical lemma, whose proof is left in the Appendix.

Lemma 7. *If G is obtained from $C_3 \square K_2$ by inserting one diamond, or from X_{12} by inserting two diamonds into edges corresponding to the underlying cubic multigraph, then $\chi_3(G) \leq 5$.*

We now proceed to prove the Hamiltonian setting.

Theorem 8. *Let G be a Hamiltonian claw-free cubic graph. Then $\chi_3(G) \leq 5$. if and only if $G \notin \{C_3 \square K_2, X_{12}, X_{16}^1, X_{16}^2\}$.*

Proof. If $G \cong K_4$, then the result is immediate. If G is a ring of diamonds, then the result follows from Observation 1. Since G is Hamiltonian, it is bridgeless. Hence, by Corollary 6, G is obtained from a loopless bridgeless cubic multigraph H by replacing every vertex of H with a triangle and every edge

of H with a possibly empty string of diamonds. Let $t_0, \ldots, t_{r-1}$ be the vertices of H, and let T_i be the triangle replacing t_i.

Let R be a Hamiltonian cycle of G. Each triangle block has exactly two edges of R leaving it, and every nonempty string of diamonds is traversed by R from one end to the other. Hence, after contracting each triangle block to a vertex and each string of diamonds to an edge, R yields a Hamiltonian cycle

$$t_0 t_1 \ldots t_{r-1} t_0$$

of H. Let F be the graph obtained from H by replacing each vertex with a triangle, or equivalently, by replacing each string of diamonds in G with a single edge. We label the vertices of T_i as a_i, b_i, c_i so that

$$D = a_0 b_0 a_1 b_1 \ldots a_{r-1} b_{r-1} a_0$$

is a cycle in F.

Since H is cubic, the edges of H not belonging to the Hamiltonian cycle form a perfect matching. Consequently, $C = \{c_0, \ldots, c_{r-1}\}$ induces a perfect matching M in F. Given a path or cycle $P = u_0 u_1 \ldots u_k$, we call a coloring ϕ of P standard if $\phi(u_i) \equiv i \pmod 3$.

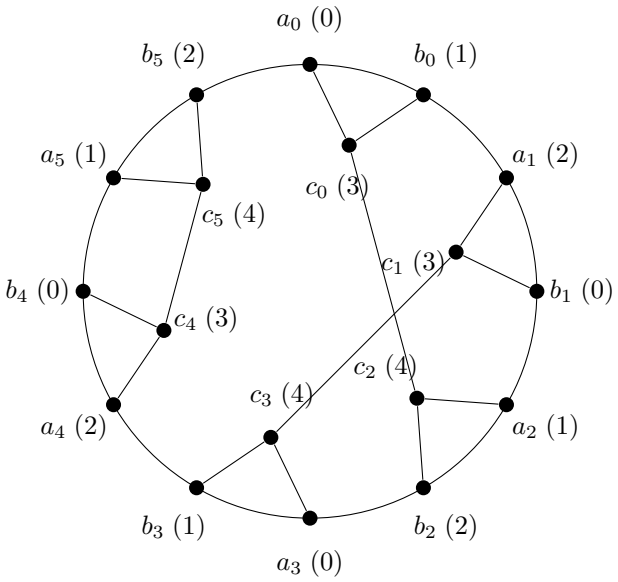


Figure 2: The case $2r \equiv 0 \pmod 3$ in the proof of Theorem 8. In this case $2r = 12$. In parentheses after each vertex label is shown the color assigned to that vertex.

Whenever $\chi_3(F) \leq 5$, repeated applications of Proposition 2 yield $\chi_3(G) \leq 5$. We now construct a 3-dynamic 5-coloring of F, except for the small configurations that will be handled separately using Lemma 7. Suppose first that $2r \equiv 0 \pmod 3$. Precolor D with a standard coloring ϕ. Since $2r$ is a multiple of 3, this coloring is well defined and proper on D. For every i (indices taken modulo r), the vertices b_{i-1} and b_i receive distinct colors, and so do a_i and a_{i+1}. Since $C = \{c_0, \ldots, c_{r-1}\}$ induces the perfect matching M, extend ϕ by assigning colors 3 and 4 to the two endpoints of each edge of M. Then a_i sees distinct colors on b_{i-1} and b_i, together with the color of c_i, and b_i sees distinct colors on a_i and a_{i+1}, together with the color of c_i. Moreover, c_i sees the two distinct colors of a_i and b_i and the color of its neighbor in M. Hence every vertex has three neighbors of pairwise distinct colors, and ϕ is a 3-dynamic 5-coloring of F (Figure 2). We may therefore assume from now on that $2r \not\equiv 0 \pmod 3$.

Suppose that every edge of M joins two consecutive vertices of C. After relabeling if necessary, we may assume that

$$M = \{c_{2i} c_{2i+1} : 0 \leq i < r/2\}.$$

We first consider the case $2r \equiv 1 \pmod 3$. Since r is even, either $r = 2$ or $r \geq 8$. If $r = 2$, then $F \cong C_3 \square K_2$. Since $\chi_3(C_3 \square K_2) = 6$, we may assume that at least one diamond occurs in G. Let F' be

obtained from F by retaining exactly one such diamond. By Lemma 7, $\chi_3(F') \leq 5$. Since G is obtained from F' by inserting the remaining diamonds, repeated applications of Proposition 2 yield $\chi_3(G) \leq 5$.

Hence, we may assume that $r \geq 8$. We precolor $a_0b_0a_1b_1 \dots a_{r-3}b_{r-3}$ with a standard coloring ϕ. We then set $\phi(c_0) = \phi(c_{r-3}) = \phi(c_{r-1}) = 3$, $\phi(c_{r-2}) = 0$, and $\phi(a_{r-2}) = \phi(b_{r-1}) = 4$, $\phi(b_{r-2}) = 1$, $\phi(a_{r-1}) = 2$. We color the remaining vertices of C with colors 3 and 4 so that the endpoints of every edge of M receive distinct colors. A direct check shows that the resulting coloring is proper and that the three neighbors of every vertex receive pairwise distinct colors. Hence ϕ is a 3-dynamic 5-coloring of F (Figure 3a).

We now consider the case $2r \equiv 2 \pmod 3$. Since r is even, $r \equiv 4 \pmod 6$. We first assume that $r \geq 10$. We precolor

$$a_0b_0a_1b_1 \dots a_{r-5}b_{r-5}$$

with a standard coloring ϕ. We then set $\phi(c_0) = \phi(c_{r-1}) = \phi(b_{r-3}) = \phi(a_{r-4}) = 3$, $\phi(a_{r-2}) = \phi(b_{r-1}) = \phi(c_{r-4}) = \phi(c_{r-5}) = 4$, $\phi(c_{r-2}) = \phi(c_{r-3}) = 0$, and $\phi(a_{r-1}) = \phi(b_{r-4}) = 1$, $\phi(b_{r-2}) = \phi(a_{r-3}) = 2$. We color the remaining vertices of C with colors 3 and 4 so that the endpoints of every edge of M receive distinct colors. A direct check shows that the resulting coloring is proper and that the three neighbors of every vertex receive pairwise distinct colors. Hence ϕ is a 3-dynamic 5-coloring of F (Figure 3b). Finally, If $r = 4$, then $M = \{c_0c_1, c_2c_3\}$, and hence $F \cong X_{12}$. If no diamond occurs, then $G \cong X_{12}$; if exactly one diamond occurs, then $G \cong X_{16}^1$ or $G \cong X_{16}^2$. It can be verified that $\chi_3(G) = 6$ for any such G. Hence, assume that at least two diamonds occur in G. Let F' be obtained from X_{12} by retaining exactly two of these diamonds. By Lemma 7, $\chi_3(F') \leq 5$. Since G is obtained from F' by inserting the remaining diamonds, repeated applications of Proposition 2 yield $\chi_3(G) \leq 5$.

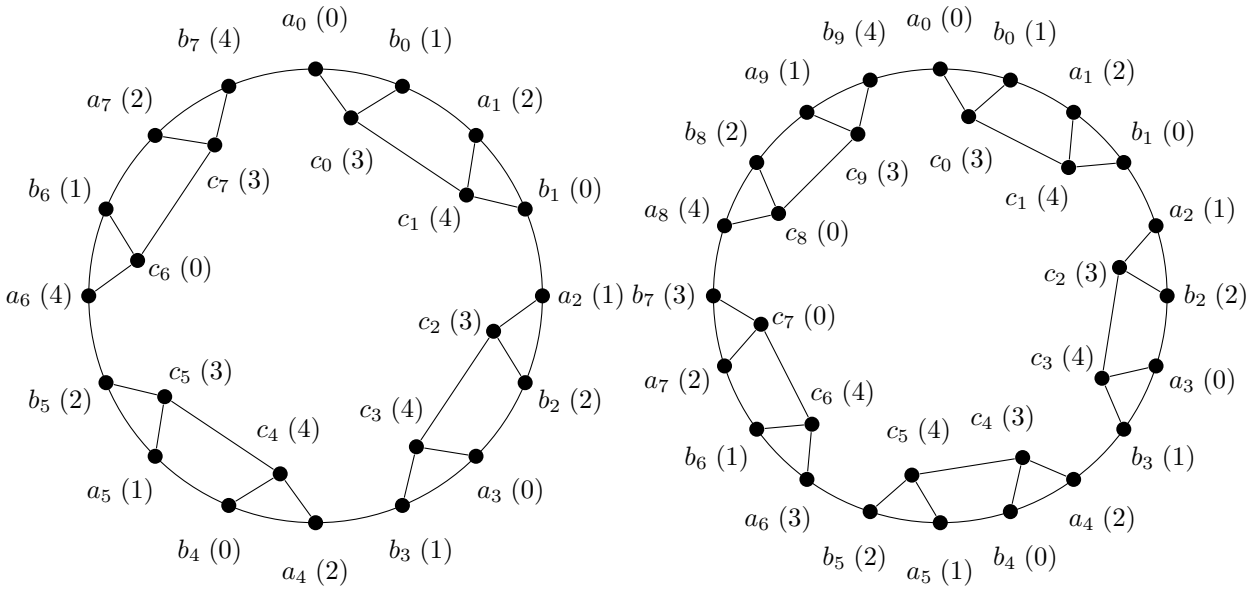


Figure 3: The case in which every edge of M joins two consecutive vertices of C in the proof of Theorem 8. In parentheses after each vertex label is the color assigned to that vertex. In (a), $2r \equiv 1 \pmod 3$ and $2r = 16$; in (b), $2r \equiv 2 \pmod 3$ and $2r = 20$.

We may therefore assume that M contains an edge joining two nonconsecutive vertices of C. Among the edges $c_ic_k \in M$ joining nonconsecutive vertices of C, choose one for which $\min\{|i-k|, r-|i-k|\}$ is maximum. After relabeling the indices and, if necessary, reversing their cyclic order, we may assume that this edge is c_0c_j, where $2 \leq j \leq r/2$. Let $P = a_1b_1a_2b_2 \dots a_{j-1}b_{j-1}$ and $Q = a_{j+1}b_{j+1}a_{j+2}b_{j+2} \dots a_{r-1}b_{r-1}$. Since $|V(P)| + |V(Q)| = 2r - 4$ and $2r \not\equiv 0 \pmod 3$, we have $|V(P)| + |V(Q)| \not\equiv 2 \pmod 3$. Moreover, since $2 \leq j \leq r-2$, both P and Q are nonempty.

Let $\overline{Q}$ denote the path Q taken in the reverse order. We precolor P and $\overline{Q}$ with standard colorings ϕ. Thus, $\phi(a_1) = \phi(b_{r-1}) = 0$. We also set $\phi(a_0) = \phi(c_1) = 3$ and $\phi(b_0) = \phi(c_{r-1}) = 4$. We claim

that c_{j-1} and c_{j+1} can be assigned distinct colors from $\{3,4\}$ while preserving properness on the already colored edges of M. If $j \geq 3$, this follows directly from the fact that M is a matching. Suppose then that $j = 2$. Since c_1 already has color 3, we only need to show that c_3 can receive color 4. The only possible obstruction is $c_3c_{r-1} \in M$. If $r \geq 8$, this contradicts the maximal choice of c_0c_2, since c_3c_{r-1} has larger cyclic distance. If $r = 4$, then $c_3 = c_{r-1}$ already has color 4.

Thus, c_{j-1} and c_{j+1} can be assigned distinct colors from $\{3,4\}$. By symmetry, we may assume that $\phi(c_{j-1}) = 3$ and $\phi(c_{j+1}) = 4$. We then set $\phi(a_j) = 4$ and $\phi(b_j) = 3$. We color every remaining vertex of $C \setminus \{c_0, c_j\}$ with colors 3 and 4 so that the endpoints of every edge of M other than c_0c_j receive distinct colors. At this point, every vertex outside $T_0 \cup T_j$ satisfies the 3-dynamic condition. It remains to color c_0 and c_j.

First suppose that one of $|V(P)|$ and $|V(Q)|$ is congruent to 0 modulo 3. By symmetry, assume that $|V(P)| \equiv 0 \pmod 3$. Since b_{j-1} is the last vertex of P, we have $\phi(b_{j-1}) = 2$. Moreover, since $|V(P)| + |V(Q)| \not\equiv 2 \pmod 3$, we have $|V(Q)| \not\equiv 2 \pmod 3$. As $\overline{Q}$ is standard-colored, it follows that $\phi(a_{j+1}) \in \{0,2\}$. We set $\phi(c_0) = 2$ and $\phi(c_j) = 1$. Since $\phi(a_0) = 3$, $\phi(b_0) = 4$, and $\phi(a_1) = \phi(b_{r-1}) = 0$, the coloring is proper and the 3-dynamic condition holds at the vertices of T_0. Furthermore,

$$\phi(b_{j-1}) = 2, \qquad \phi(a_j) = 4, \qquad \phi(b_j) = 3, \qquad \phi(a_{j+1}) \in \{0,2\}.$$

Hence the coloring is proper and the 3-dynamic condition also holds at the vertices of T_j. Therefore, ϕ is a 3-dynamic 5-coloring of F (Figure 4a).

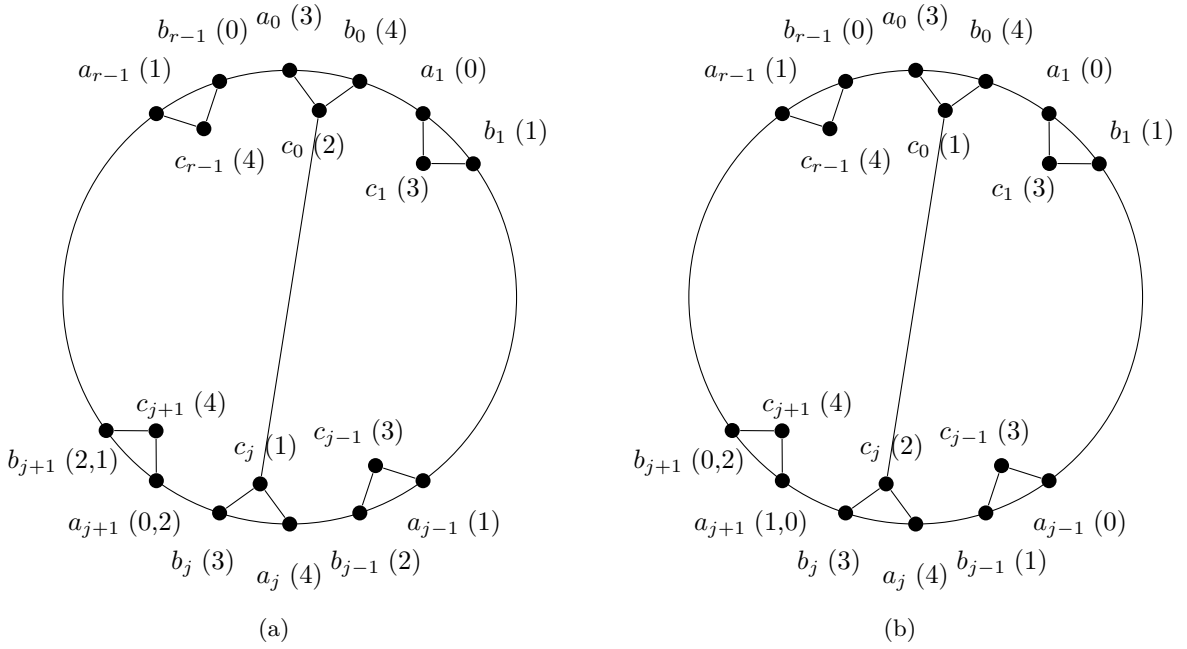


Figure 4: Situation in the proof of Theorem 8. In this case, there exist an edge c_0c_j with $j > 0$. In parenthesis after each vertex label is shown the color assigned to such vertex. In (a) we depict the case $|V(P)| \equiv 0 \pmod 3$. In (b) we depict the case $|V(P)| \equiv 2 \pmod 3$.

We may therefore assume that neither $|V(P)|$ nor $|V(Q)|$ is congruent to 0 modulo 3. Since $|V(P)| + |V(Q)| \not\equiv 2 \pmod 3$, at least one of them is congruent to 2 modulo 3. By symmetry, assume that $|V(P)| \equiv 2 \pmod 3$. Then $\phi(b_{j-1}) = 1$. Moreover, $|V(Q)| \not\equiv 0 \pmod 3$, and, since $\overline{Q}$ is standard-colored, $\phi(a_{j+1}) \in \{0,1\}$. We set $\phi(c_0) = 1$ and $\phi(c_j) = 2$. Again, since $\phi(a_0) = 3$, $\phi(b_0) = 4$, and $\phi(a_1) = \phi(b_{r-1}) = 0$, the coloring is proper and the 3-dynamic condition holds at the vertices of T_0. Moreover,

$$\phi(b_{j-1}) = 1, \qquad \phi(a_j) = 4, \qquad \phi(b_j) = 3, \qquad \phi(a_{j+1}) \in \{0,1\}.$$

Thus the coloring is proper and the 3-dynamic condition also holds at the vertices of T_j. Therefore, ϕ is a 3-dynamic 5-coloring of F (Figure 4b). □

3 2-dynamic coloring of $C_p(1,3)$

We use color words cyclically on the vertices $0, 1, \ldots, p-1$; as usual, W^q denotes the concatenation of q copies of a word W.

Theorem 9. *Let $p \geq 6$. Then*

$$\chi_2^d(C_p(1,3)) = \begin{cases} 3, & p \equiv 0 \pmod 9, \\ 4, & \textit{otherwise.} \end{cases}$$

Proof. If $9 \mid p$, the word $(012120201)^{p/9}$ defines a 2-dynamic 3-coloring: each vertex has a color different from its neighbors at distances 1 and 3, and its neighborhood contains both other colors. Hence $\chi_2^d(C_p(1,3)) = 3$ by Equation 1.

We next give a 2-dynamic 4-coloring for every $p \geq 6$. Let $A = 0123$ and write $p = 4q + r$, $0 \leq r \leq 3$. Use the following cyclic color words:

r	0	1	2	3
ϕ	A^q	$A^{q-1}03231$	A^q01	A^q201

A direct verification shows that ϕ is a 2-dynamic coloring. Thus $\chi_2^d(C_p(1,3)) \leq 4$ for every $p \geq 6$.

It remains to prove that a 2-dynamic 3-coloring can exist only if $9 \mid p$. We may assume $p \geq 7$, since $C_6(1,3) \cong K_{3,3}$ has no 2-dynamic 3-coloring. Let ϕ be a 2-dynamic 3-coloring and define

$$s_i = \begin{cases} 1, & \phi(i-1) \neq \phi(i+1), \\ 0, & \phi(i-1) = \phi(i+1). \end{cases}$$

We first observe that $s_i = s_{i+1} = 1$ is impossible. Indeed, in this case both $\phi(i-1), \phi(i), \phi(i+1)$ and $\phi(i), \phi(i+1), \phi(i+2)$ contain the three colors. Hence $\phi(i-1) = \phi(i+2)$, contradicting the fact that $i-1$ and $i+2$ are adjacent.

Moreover, $s_{i-2} = s_i = s_{i+2} = 0$ is impossible, since it would imply $\phi(i-3) = \phi(i-1) = \phi(i+1) = \phi(i+3)$, contrary to the 2-dynamic condition at i.

Since not all s_i can be zero, choose i with $s_i = 1$. Then $s_{i-1} = s_{i+1} = 0$, and the previous observation forces $s_{i-3} = s_{i+3} = 1$. By the first observation, $s_{i-2} = s_{i+2} = 0$. Reconstructing ϕ yields $\phi(i-4) = \phi(i+5)$. Repeating the same argument gives

$$\phi(i-4) = \phi(i-4+9k)$$

for every k, and hence $p \equiv 0 \pmod 9$. □

4 3-dynamic coloring of $C_p(1,3)$

Theorem 10. *Let $p \geq 12$. Then*

$$\chi_3^d(C_p(1,3)) = \begin{cases} 4, & p \equiv 0 \pmod{12}, \\ 5, & \textit{otherwise.} \end{cases}$$

Proof. Suppose first that $12 \mid p$. The cyclic color word $(010232101323)^{p/12}$ defines a proper 3-dynamic 4-coloring. Hence $\chi_3^d(C_p(1,3)) = 4$ by Equation 1.

We next give a 3-dynamic 5-coloring for every $p \geq 12$. Let $A = 01234$ and write $p = 5q + r$, where $0 \leq r \leq 4$. Consider the following cyclic color words:

r	0	1	2	3	4
ϕ	A^q	$A^{q-1}102431$	A^q01	$A^{q-1}01314203$	A^q1203

A direct verification shows that each word defines a proper 3-dynamic 5-coloring. Thus $\chi_3^d(C_p(1,3)) \leq 5$ for every $p \geq 12$.

It remains to show that a 3-dynamic 4-coloring can exist only if $12 \mid p$. Let ϕ be such a coloring. Since ϕ is proper, the neighbors of every vertex use exactly the three colors different from the color of the vertex. Thus exactly two of its four neighbors receive the same color.

Define

$$s_i = \begin{cases} 1, & \phi(i-1) = \phi(i+1), \\ 0, & \phi(i-1) \neq \phi(i+1). \end{cases}$$

We first observe that the repeated color in $N(i)$ can only occur on one of the pairs $\{i-3, i-1\}$, $\{i-1, i+1\}$, $\{i+1, i+3\}$. Indeed, if $\phi(i-3) = \phi(i+1)$, then properness forces $\phi(i+2) = \phi(i)$, and the 3-dynamic condition at $i-1$ forces one of $i-4, i-2$ to receive the color of $i-3$, contradicting properness. The case $\phi(i-1) = \phi(i+3)$ is symmetric. If $\phi(i-3) = \phi(i+3)$, then properness forces $\phi(i-2) = \phi(i+2) = \phi(i)$, contradicting the 3-dynamic condition at $i-1$.

Consequently, for every i, exactly one of

$$s_{i-2},\ s_i,\ s_{i+2}$$

is equal to 1. Moreover, two consecutive s_i cannot both be 1. Indeed, if $s_i = s_{i+1} = 1$, a direct reconstruction using the 3-dynamic condition around i and $i+1$ forces $\phi(i+4) = \phi(i+5)$, contradicting properness.

Choose i with $s_i = 1$. Then by the previous observations, $s_{i-6} = s_{i+6} = s_{i-3} = s_{i+3} = 1$ and every other s_k between $i-6$ and $i+6$ equals 0. This implies $3 \mid p$. Reconstructing ϕ yields $\phi(i-6) = \phi(i+6)$. Repeating the same argument gives

$$\phi(i-6) = \phi(i-6+12k) \neq \phi(i)$$

for every k, and hence, as $p \equiv 0 \pmod{12}$. □

5 4-dynamic coloring of $C_p(1,3)$

Theorem 11. *Let $p \geq 12$. Then*

$$\chi_4^d(C_p(1,3)) = \begin{cases} 5, & p \equiv 0 \pmod 5, \\ 8, & p = 16, \\ 7, & p \in \{13, 14, 19, 21, 26, 31\}, \\ 6, & \textit{otherwise.} \end{cases}$$

Proof. Since $C_p(1,3)$ is 4-regular, a proper coloring is 4-dynamic if and only if vertices at distance at most two receive different colors. Hence

$$\chi_4^d(C_p(1,3)) = \chi(H_p),$$

where $H_p = C_p(1,2,3,4,6)$. For every i, the set $\{i, i+1, i+2, i+3, i+4\}$ induces a K_5 in H_p. Thus $\chi(H_p) \geq 5$. Moreover, in any 5-coloring, two consecutive such cliques force $\phi(i+5) = \phi(i)$ for every i. Therefore a 5-coloring can exist only if $5 \mid p$. Consequently, $\chi(H_p) \geq 6$ whenever $5 \nmid p$.

We need stronger lower bounds only for a few small values of p. Let S be an independent set of H_p, and let $g_1, \ldots, g_t$ be the cyclic distances between consecutive vertices of S. Since the distances $1, 2, 3, 4, 6$ are forbidden, every g_j is either 5 or at least 7. Hence

$$\alpha(H_p) \leq \left\lfloor \frac{p}{5} \right\rfloor.$$

Furthermore, if $p = 5q + 1$, then $\alpha(H_p) \leq q - 1$, since q distances would require $1 = \sum_{j=1}^{q}(g_j - 5)$, where every summand is either 0 or at least 2. Using $\chi(H_p) \geq \left\lceil \frac{p}{\alpha(H_p)} \right\rceil$, we obtain $\chi(H_{16}) \geq 8$ and $\chi(H_p) \geq 7$ for $p \in \{13, 14, 19, 21, 26, 31\}$.

It remains to give matching colorings. Let

$$A = 01234, \qquad B = 012340521435, \qquad C = 012340153421504235.$$

If $p = 5q$, then A^q is a proper 5-coloring of H_p.

For the nonexceptional values with $5 \nmid p$, write $p = 5q + r$. The following cyclic color words use six colors:

r	condition	ϕ
1	$q \geq 7$	$A^{q-7}C^2$
2	$q \geq 2$	$A^{q-2}B$
3	$q \geq 3$	$A^{q-3}C$
4	$q \geq 4$	$A^{q-4}B^2$.

A direct verification shows that each word is a proper coloring of H_p.

Finally, let

$$D = 0123456, \qquad E = 01234567.$$

The exceptional cases are covered by

p	ϕ
13	$D201534$
14	D^2
16	E^2
19	D^2A
21	D^3
26	D^3A
31	$D^4\,234$.

Again, a direct verification shows that these are proper colorings of H_p. They use seven colors, except for $p = 16$, where eight colors are used. Together with the lower bounds above, the result follows. □

6 Concluding remarks

In this paper, we investigated r-dynamic colorings of regular graphs for claw-free cubic graphs and circulant graphs. For cubic graphs, 3-dynamic coloring coincides with coloring the square of the graph. This connection allows the problem to be viewed from the perspective of square coloring while exploiting the strong structural properties of claw-free cubic graphs. We proved that every claw-free cubic graph admits a 3-dynamic coloring with at most six colors, and we identified the extremal cases showing that this bound is best possible. For Hamiltonian claw-free cubic graphs, the additional structure permits a stronger bound of five colors, apart from the exceptional graphs $C_3 \square K_2$, X_{12}, X_{16}^1, and X_{16}^2.

The circulant case illustrates a substantially different phenomenon. Here the parameter r is smaller than the degree, and consequently r-dynamic coloring no longer reduces to coloring a graph square. For the circulants considered in this work, this distinction allows considerably more economical colorings and leads to exact results depending on the arithmetic structure of the underlying cycle.

Several questions remain open. A natural starting point is the general situation for r-regular graphs. Since an r-dynamic coloring of an r-regular graph is precisely a proper coloring of its square, one always has $r + 1 \leq \chi_r(G) = \chi(G^2) \leq r^2 + 1$. The upper bound follows immediately from $\Delta(G^2) \leq r^2$, and it is best possible in general, as shown by the classical Moore graph examples. Thus, even in the regular setting, the possible range for the r-dynamic chromatic number is quite large.

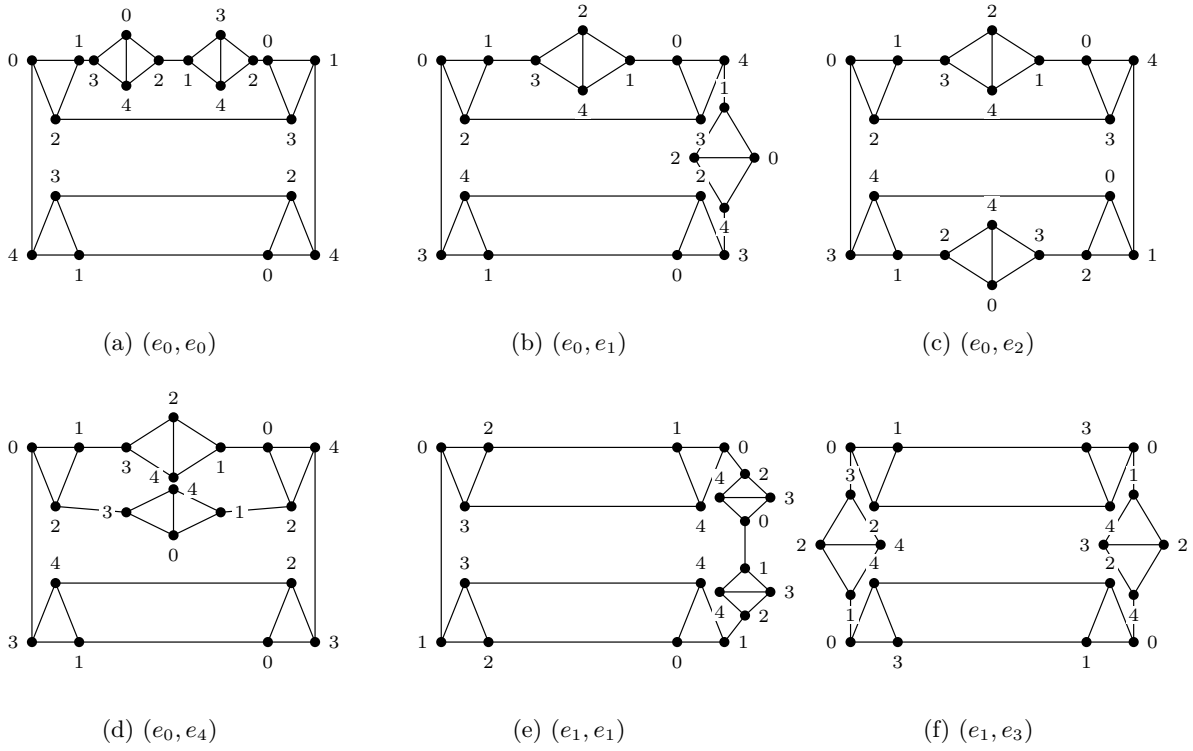
For cubic graphs, this general estimate gives $4 \leq \chi_3(G) \leq 10$, and the upper bound cannot be improved over the class of all cubic graphs, since the Petersen graph satisfies $\chi_3(G) = 10$. From this perspective, the bounds obtained here show that forbidding an induced claw imposes a substantial restriction on the square coloring problem: every claw-free cubic graph in our setting requires at most six colors, and under the additional Hamiltonicity assumption this can be reduced further to five, apart from the exceptional configurations identified above. It would therefore be interesting to determine which structural conditions on cubic or, more generally, regular graphs force the r-dynamic chromatic number to lie substantially below the general bound $r^2 + 1$.

7 Appendix: Proof of Lemma 7.

Proof. For the first part, let the two triangles of the prism be $a_0b_0c_0$ and $a_1b_1c_1$, and suppose without loss of generality that c_0c_1 is replaced by a diamond. Let p and q be the degree-2 vertices of the diamond, where p is adjacent to c_0 and q is adjacent to c_1, and let x and y be its other two vertices. Define $\phi(c_0) = \phi(c_1) = 0$, $\phi(a_0) = \phi(q) = 1$, $\phi(b_0) = \phi(x) = 2$, and $\phi(a_1) = \phi(p) = 3$, $\phi(b_1) = \phi(y) = 4$. A direct check shows that ϕ is proper and that the three neighbors of every vertex receive pairwise distinct colors. Hence ϕ is a 3-dynamic 5-coloring.

For the second part, let the four triangles of X_{12} be $\{a_ib_ic_i : i \in \{0,1,2,3\}\}$ and $e_0 = b_0a_1$, $e_1 = b_1a_2$, $e_2 = b_2a_3$, $e_3 = b_3a_0$, $e_4 = c_0c_1$, $e_5 = c_2c_3$ be the their edges outside triangles. Up to automorphisms of X_{12}, the possible placements of two diamonds are (e_0, e_0), (e_0, e_1), (e_0, e_2), (e_0, e_4), (e_1, e_1), (e_1, e_3).

There are six configurations, shown in Figure 5; the displayed colorings verify the claim. □



(a) (e_0, e_0) (b) (e_0, e_1) (c) (e_0, e_2)

(d) (e_0, e_4) (e) (e_1, e_1) (f) (e_1, e_3)

Figure 5: The six configurations, up to symmetry, obtained from X_{12} by inserting two diamonds. The number shown next to each vertex is its assigned color.

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